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-- 1. DDL for actors table
create type film_role as
(
    filmid text,
    film    text,
    votes   integer,
    rating  real
);

create type quality_class as enum ('star', 'good', 'average', 'bad');

create table actors
(
    actor          text,
    actorid        text,
    films          film_role[]    not null,
    quality_class  quality_class not null,
    is_active      boolean,
    current_year   integer,
    primary key (actorid, current_year)
);

create index idx_actors_current_year
on actors (current_year);

create index idx_actors_actorid_current_year
on actors (actorid, current_year);
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-- 2. Cumulative table generation query

-- Let's create an index on actor_films first
create index idx_actor_films_actor_year_desc
on actor_films (actorid, year desc);

-- Fill the table
insert into actors
with years as (select *
                from generate_series((select min(year)
                                       from actor_films),
                                     (select max(year)
                                       from actor_films)) as year),
actor_film_as_of_year as (select af.*,
                                y.year as current_year
                                from years y
                                join actor_films af
                                on af.year <= y.year),
actor_films_as_of_year_agg as (select actorid as
actorid,
                                array_agg(row (
                                afaoy.filmid,
                                afaoy.film,
                                afaoy.votes,
                                afaoy.rating)::film_role
                                order by filmid) as
films,
                                max(afaoy.year) = current_year as
is_active,
                                current_year
                                from actor_film_as_of_year afaoy
                                group by current_year, actorid),
actor_filming_year_quality_class as (select current_year,
actorid,
                                year
                                as films_year,
                                case
                                when avg(rating) > 8
                                then 'star'
                                when avg(rating) > 7
                                then 'good'
                                when avg(rating) > 6
                                then 'average'
                                else 'bad'
                                end::quality_class as quality_class
                                from actor_film_as_of_year afaoy
                                group by current_year, actorid,
                                afaoy.year
                                order by 1, 2, 3 desc),
actor_last_active_quality_class as (select distinct on (current_year,
actorid) current_year,
actorid,
quality_class

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                                from
actor_filming_year_quality_class),
    actor_films_cumulated as (select distinct on (afaoya.actorid,
afaoya.current_year) afaoya.actorid,

afaoya.films,

alaqc.quality_class,

afaoya.is_active,

afaoya.current_year

                                from actor_films_as_of_year_agg afaoya
                                join actor_last_active_quality_class
alaqc
                                on afaoya.actorid =
alaqc.actorid and
                                afaoya.current_year =
alaqc.current_year),
    actor_dim as (select distinct on (actorid) actorid, actor
                    from actor_films
                    where actor is not null)
select dim.actor as actor,
    afc.actorid,
    afc.films,
    afc.quality_class,
    afc.is_active,
    afc.current_year
from actor_films_cumulated afc
    join actor_dim dim on afc.actorid = dim.actorid
;

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-- 3. DDL for actors_history_scd table
create table actors_history_scd
(
    actorid      text,
    actor        text,
    quality_class quality_class,
    is_active    boolean,
    start_year   integer,
    end_year     integer,
    current_year integer,
    primary key (actorid, start_year, current_year)
);
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-- 4. Backfill query for actors_history_scd
DO
$$
    DECLARE
        start_year integer := (select min(current_year)
                                from actors);
        end_year   integer := (select max(current_year)
                                from actors);
    BEGIN
        FOR curr_year IN start_year..end_year
            LOOP
                insert into actors_history_scd
                with with_previous as (select actorid,
                                              actor,
                                              current_year,
                                              quality_class,
                                              is_active,
                                              lag(quality_class, 1)
                                              over (partition by actorid
order by current_year) as prev_quality_class,
                                              lag(is_active, 1)
                                              over (partition by actorid
order by current_year) as prev_is_active
                                              from actors
                                              where current_year <= curr_year),
                    with_indicators as (select *,
                                              case
prev_quality_class then 1
prev_is_active then 1
                                              else 0
                                              end as change_indicator
                                              from with_previous),
                    with_streaks as (select *,
                                      sum(change_indicator)
                                      over (partition by actorid order
by current_year) as streak_identifier
                                              from with_indicators)
                select actorid,
                       min(actor),
                       quality_class,
                       is_active,
                       min(current_year) as start_year,
                       max(current_year) as end_year,
                       curr_year         as current_year
                from with_streaks
                group by actorid, streak_identifier, quality_class,
is_active;
            END LOOP;
        END
    $$;

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-- 5. Incremental query for actors_history_scd

-- Clean actors_history_scd
truncate actors_history_scd;

-- Backfill scd until 1981
DO
$$
    DECLARE
        start_year integer := (select min(current_year)
                                from actors);
        end_year   integer := (select 1981);
    BEGIN
        FOR curr_year IN start_year..end_year
        LOOP
            insert into actors_history_scd
            with with_previous as (select actorid,
                                         actor,
                                         current_year,
                                         quality_class,
                                         is_active,
                                         lag(quality_class, 1)
                                         over (partition by actorid
order by current_year) as prev_quality_class,
                                         lag(is_active, 1)
                                         over (partition by actorid
order by current_year) as prev_is_active
                                         from actors
                                         where current_year <= curr_year),
            with_indicators as (select *,
                                     case
prev_quality_class then 1
prev_is_active then 1
                                     when quality_class <>
                                     when is_active <>
                                     else 0
                                     end as change_indicator
                                     from with_previous),
            with_streaks as (select *,
                                sum(change_indicator)
                                over (partition by actorid order
by current_year) as streak_identifier
                                from with_indicators)
            select actorid,
                   actor,
                   quality_class,
                   is_active,
                   min(current_year) as start_year,
                   max(current_year) as end_year,
                   curr_year         as current_year
            from with_streaks
            group by actorid, streak_identifier, actor, quality_class,
is_active;
        END LOOP;
    END
$$;

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-- Get scd records number per year
select current_year, count(*) as num_records
from actors_history_scd
group by current_year
order by 1 desc;

-- Get scd records as of last_year
with params as (select max(current_year) as last_scd_year from
actors_history_scd)
select scd.*
from actors_history_scd scd
      join params p
            on scd.current_year = p.last_scd_year
order by actorid, start_year
;

-- Fill with new actors data as of 1982
insert into actors_history_scd
with params as (select 1982 as curr_year),
      scd_as_of_last_year as (select scd.*
                                from actors_history_scd scd
                                join params p
                                on scd.current_year = p.curr_year
- 1),
      scd_ongoing_streaks_as_of_last_year as (select actorid,
                                                    actor,
                                                    quality_class,
                                                    is_active,
                                                    start_year,
                                                    end_year
                                                    from scd_as_of_last_year
                                                    where end_year = current_year),
      this_year_data as (select a.*
                            from actors a
                            join params p
                            on a.current_year = p.curr_year),
      unchanged_scd as (select td.actorid,
                                td.actor,
                                td.quality_class,
                                td.is_active,
                                ls.start_year,
                                td.current_year
                                from this_year_data td
                                join scd_ongoing_streaks_as_of_last_year ls
                                on td.actorid = ls.actorid and
td.current_year = ls.end_year + 1
                                where td.quality_class = ls.quality_class
                                and td.is_active = ls.is_active),
      changed_scd as (select td.actorid,
                                td.actor,
                                td.quality_class,
                                td.is_active,
                                td.current_year,
                                td.current_year
                                from this_year_data td
                                join scd_ongoing_streaks_as_of_last_year ls

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                                on td.actorid = ls.actorid and
td.current_year = ls.end_year + 1
                                where td.quality_class <> ls.quality_class
                                    or td.is_active <> ls.is_active),
    history_scd as (select actorid,
                           actor,
                           quality_class,
                           is_active,
                           start_year,
                           end_year
                    from scd_as_of_last_year scd
                    where not exists(select 1
                                     from unchanged_scd uscd
                                     where scd.actorid = uscd.actorid
                                     and scd.start_year =
uscd.start_year)),
    new_scd as (select td.actorid,
                       td.actor,
                       td.quality_class,
                       td.is_active,
                       td.current_year,
                       td.current_year
                from this_year_data td
                left join scd_ongoing_streaks_as_of_last_year ls
                    on td.actorid = ls.actorid and
td.current_year = ls.end_year + 1
                    where ls.actorid is null),
    accumulated_scd as (select *, 'history' as source
                        from history_scd
                        union all
                        select *, 'unchanged' as source
                        from unchanged_scd
                        union all
                        select *, 'changed' as source
                        from changed_scd
                        union all
                        select *, 'new' as source
                        from new_scd)
select acc.actorid,
       acc.actor,
       acc.quality_class,
       acc.is_active,
       acc.start_year,
       acc.end_year,
       p.curr_year as current_year
from accumulated_scd acc,
     params p
;

-- Get scd records number per year
select current_year, count(*) as num_records
from actors_history_scd
group by current_year
order by 1 desc;

-- Get scd records as of last_year
with params as (select max(current_year) as last_scd_year from

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actors_history_scd)
select scd.*
from actors_history_scd scd
      join params p
        on scd.current_year = p.last_scd_year
order by actorid, start_year
;
```